[13] Single-linkage clustering (SLC) is a class of bottom-up hierarchical clustering algorithms where, at each step, the pair of clusters at minimum distance is merged (initially every point is in its own cluster).

[14] *Reductio ad absurdum*, from Latin “reduction to absurdity” is a logic argument that proves a statement by showing that, if false, it would lead to an impossible result.

[15] Some of the neighbors of the core point could have already been merged.

[16] As we have seen in the previous sections, the cost function that k-means minimizes is the Euclidean distance to centroids (and, indirectly, the in-cluster Euclidean distance).

[17] The running time of these queries, as we have seen in chapter 9, can’t be upper-bounded by anything better than `O(n)` in the worst case. However, the running time of range queries on hyper-spheres depends on the radius of the sphere, and under certain assumptions on the value of ε, `O(log(n))` can be an accurate estimate for the average running time.

[18] In machine learning, the parameters passed to an algorithm, and fixed before learning starts, are often called hyperparameters to distinguish them from the parameters of the model (for instance, the weights of a neural network) produced by the algorithm.

[19] Ankerst, Mihael, et al. "OPTICS: ordering points to identify the clustering structure." ACM Sigmod record. Vol. 28. No. 2. ACM, 1999.

[20] Here we adopt the definition of “neighbor of a cluster” meaning a point that is a neighbor (and hence reachable) of any point currently in the cluster.

[21] Assuming we consider `p` as its own neighbor… technically, since we agreed that for `p` to be a core point there needs to be `M` points within a radius ε from `p`, to compute the core distance we take the distance of its `(M-1)-th` neighbor.

[22] Technically, as we have noted, it simultaneously grows an infinite number of clusters, for all the possible values of density allowed – but for the sake of simplicity, we will focus on a single cluster in the description, the one that would be obtained with $\varepsilon=\varepsilon_{MAX}$.

[23] Also, by using a Fibonacci heap, the “update priority” method, lowering priorities, would only require an amortized `O(1)` running time, so even a linearithmic number of calls would at most require `O(n*log(n))` total time.

[24] Achtert, Elke, Christian Böhm, and Peer Kröger. "DeLi-Clu: boosting robustness, completeness, usability, and efficiency of hierarchical clustering by a closest pair ranking." Pacific-Asia Conference on Knowledge Discovery and Data Mining. Springer, Berlin, Heidelberg, 2006.

[25] This is called z-score normalization, and has the advantage to produce data with zero-mean and unit-variance. There are other ways to perform feature scaling, to learn more take a look, for instance, at chapter 2 of “Machine Learning in Action”.

[26] If you are optimizing the choice for current dataset, picking the best result makes sense; if you’ll have to run the algorithm on several datasets with similar characteristics, then you might need to find a value that can generalize well, and using the median could be your best bet.